

RITIK KUMAR

5th-year, BS+MS (Research) Physics | Indian Institute of Science, Bengaluru
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EDUCATION

Indian Institute of Science, Bengaluru, India.
(BS+MS) in Physics

Aug. 2022 – July 2027 (expected)
CGPA: 7.5/10.0

- PhD-level coursework via Joint Astronomy Programme (IISc, IIA, RRI, ISRO): Fundamental of Astrophysics, Radiative Processes, Galaxies & ISM, General Relativity & Cosmology, Astronomical Techniques, Fluid Mechanics & Plasma Physics
- Linear Algebra, Multi-variable Calculus, Probability & Statistics, Machine Learning, Matrix Theory, Stochastic models & Application, Mathematical Methods in Physics, Computational Physics.

RESEARCH EXPERIENCE

Spectral Analysis of Seyfert Galaxies

Dec. 2024 – Jan. 2026

Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru

Conducted this independent research project in collaboration with Prof. C. S. Stalin.

- Developed a Python pipeline to cross-match NLSy1/BLSy1 catalogs with NuSTAR observations (12" radius) using angular-separation and signal-to-noise criteria.
- Reduced NuSTAR data with NuSTARDAS (FPMA/FPMB), including calibration, background filtering, and barycentric corrections.
- Performed broadband spectral modeling in XSPEC using cutoffpl, nthcomp, pexrav, xillverCp, and relxill to constrain coronal temperature, high-energy cutoff, and reflection properties.
- Performed spectral and timing-variability analyses across multiple epochs, generating energy-resolved light curves and calculating fractional variability with uncertainties.
- Measured coronal temperatures across multiple epochs with well-constrained uncertainties, finding evidence for temporal stability in several sources.
- **Reported coronal temperature measurements for multiple sources and investigated coronal geometry, thickness, reflection fraction, and variability. In one source, multi-epoch observations revealed significant coronal-temperature variability.**
- *Manuscript is in preparation.*

Tracing Outflows Across Different Gas Phases in AGN

Sep. 2025 – Apr. 2026

Bachelor's Thesis

Advisor: **Prof. C. S. Stalin**, IIA, Bengaluru

- Systematically investigated ionized outflows in AGN by cross-correlating the Million Quasars (MilliQuas) catalog with archival XMM-Newton observations using a 5" matching radius.
- Investigated connections between different outflow phases and their characteristic spatial and physical scales using X-ray observations.
- Identified several AGN lacking detailed outflow studies and performed XMM-Newton spectral modelling, finding multiple outflow components with velocities of 0.06–0.33c "UFO-Forest" in several systems.
- Modeling methodology: Began with a simple power-law continuum, then incorporated reflection components to isolate residual features associated with outflows. After identifying line-like signatures, Gaussian absorption/emission models (gabs, zgauss) were applied for detection. Finally, physical modeling was performed using zxipcf and XSTAR grids for the final spectral analysis.
- Performed spectral and timing-variability analyses, including energy-resolved light curves and fractional-variability measurements with uncertainties.
- We have found the sources which are showing the warm absorption and UFO from Fe K line. And interesting thing is, we have found 2 to 3 source for the first time and these sources have soft x-ray ultrafast outflows like the paper by J N Reeves found in PDS 456.
- Developed custom XSTAR photoionisation grids to model soft X-ray absorption features and Fe K-line absorption features that could not be adequately described by the available zxipcf model.

- **Manuscript is in preparation.** on a multi-phase outflow system exhibiting both soft X-ray and Fe K ultra-fast outflow signatures, with measured velocities of 0.22c and 0.32c, respectively, using custom XSTAR photoionisation grids.

Lead Researcher – Relativistic Reflection Modeling in AGN

Jan. 2026 – Aug. 2026

Full Member, ECAP X-ray Research Group

Collaboration with **Prof. Jörn Wilms, Dr. Thomas Dauser, , Alexey Nekrasov**

Erlangen Centre for Astroparticle Physics (ECAP), Friedrich-Alexander-Universität, Germany

- Leading an independent research project on developing and testing new relativistic reflection (`relxill_ring`) models for AGN X-ray spectroscopy.
- **Granted full membership in the ECAP X-ray research group (Jan 2026)**, including:
 - * Full SSH access to Remeis Observatory computing cluster for large-scale data analysis, Active membership in Sternwarte X-ray research mailing list, Regular participation in weekly ECAP X-ray group research meetings
- Led the observational analysis, including complete XMM-Newton and NuSTAR data-reduction pipelines, calibration, filtering, spectral extraction, broadband spectral modelling, and MCMC-based parameter estimation.
- Tested and validated new relativistic reflection model variants (extended corona, ring-like corona given by [Nekrasov, A. D, et al. 2025](#)), with real observational data, analyzing reflection signatures, disk-corona geometry, height of corona, and spin measurements through detailed XSPEC/ISIS modeling. Monte Carlo performed to check the parameter distribution.
- Found a boost parameter of approximately 1.2, indicating that the inferred coronal geometry deviates from a simple ring configuration.
- **Manuscript is in Preparation.**

Master’s Thesis: Constraining AGN Coronal Geometry using X-ray Polarimetry and Relativistic Reflection

Aug. 2026 – Present

Advisors: **Dr. Thomas Dauser** and **Prof. Jörn Wilms**, (ECAP) Friedrich–Alexander University Erlangen–Nürnberg (FAU), Germany

- Investigating the geometry of the X-ray corona in Active Galactic Nuclei (AGN) through a joint analysis of *IXPE*, *XMM-Newton*, and *NuSTAR* observations.
- Modeling relativistic X-ray reflection using the `relxill_ring` framework to test whether coronal geometries inferred from X-ray polarimetry are consistent with broadband X-ray reflection spectra.
- Performing a detailed study of NGC 4151 by combining X-ray polarimetry and relativistic reflection modeling to constrain the physical properties of the corona.
- Extending the `relxill` framework to investigate physically motivated extended coronal geometries and compare their observational signatures with existing models.

ML Researcher – GRB Light Curve Reconstruction

Apr 2024 – July 2026

National Astronomical Observatory of Japan (NAOJ) Advisor: Prof. Maria G. Dainotti

- Developed and evaluated multiple deep learning models in PyTorch, including ReFANN, Siamese Neural Networks, and Deep Gaussian Processes, for uncertainty-aware reconstruction of irregularly sampled Gamma-Ray Burst X-ray light curves.
- **Siamese network:** shared encoder, four-component fusion (query, reference, difference, Hadamard product $\rightarrow$ 128-dim), MC Dropout uncertainty ($p=0.1$, Gal & Ghahramani 2016), physics-informed composite loss (Gaussian NLL + $\lambda=10^{-3}$ monotonicity penalty).
- **ReFANN pipeline:** 522/545 GRBs processed (98.2%), 62% mean parameter uncertainty reduction vs. 39% benchmark, test MSE 0.0196; 5-fold CV, bootstrap resampling, z-score normalisation in log-space.
- **Deep Gaussian Process (DGP):** Implemented a two-layer stacked GP (GPyTorch, RBF kernel, Cholesky variational strategy, 35/25 inducing points) applied to 521/545 GRBs; achieved 11.6%, 12.3%, and 15.9% uncertainty reduction in $\log T_a$, $\log F_a$, and α respectively with α outlier rate of 1.92% — competitive with BNN while requiring no Monte Carlo weight sampling.
- Implemented MC Dropout convergence analysis ($n_{\text{samples}}=200$) and ablation studies across embedding dims (16/32/64) to empirically validate design choices.
- Applied robust statistics: IQR/1.35-based σ estimation, 3- σ MAD outlier rejection, spatial-correlation bootstrap ($\sigma=2.0$), 95% CI reporting across 544 GRBs.

- **Published:** 1. Kaushal, Dainotti, Ritik et al. (2026), and 2. Ayush, Ritik, Dainotti et al. (2026-arxiv pre-print and it is in (peer-review in journal). 3. M G Dainotti,,Ritik et al.(2024), Code publicly available: [Github](#).

Physics-Informed Neural Networks for ODEs/PDEs

Jan. 2025 – Apr. 2025

Advisor: **Prof. Rishita Das**, IISc, Bengaluru

- Implemented PINNs in PyTorch to capture discontinuities in shock tubes.
- Compared PINNs with traditional numerical methods (Runge-Kutta, Euler Method, central difference), which proved more reliable for handling discontinuities.
- Solved 1D/2D/3D Burgers' equations using different initial and boundary conditions, Poisson equation. We matched our results of PINNs with Numerical method and they were nearly the same.
- Solved the 1D Euler's equation using PINNs with initial Sod Shock Tube, and we captured partial shocks.
- Applied CFL stability criterion ($CFL = 0.5$) with 400 grid points for stable time-stepping in compressible flow simulations with shock wave propagation.

PUBLICATIONS & PRESENTATIONS

Publications

- Ayush, **Ritik**, Dainotti et al. (2026). Physics-Informed Neural Networks and Data-Driven Models for GRB X-ray Light-Curve Gap Reconstruction. 2026. *arXiv pre-print, paper is in peer-review*.
- Kaushal, A., Dainotti, M.G., **Ritik**, et al. (2026). Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves. *Journal of High Energy Astrophysics (JHEAp)*. 2026
- Dainotti,,**Ritik** et al. (2024). An optical gamma-ray burst catalogue with measured redshift – I. Data release of 535 gamma-ray bursts and colour evolution. *MNRAS Journal*. 2024

In Preparation

- **Ritik**, Stalin, C.S., Reeves, J.N. Possible detection of unambiguous outflows in a Luminous Quasar. In preparation, 2026.
- **Ritik**, Stalin, C.S., Hota, J. Variation in Coronal Temperature in Seyfert-type AGN. In preparation, 2026.
- Testing and Constraining the coronal geometry using the new Relativistic model.

Conference Presentations

- **XRISM International Conference 2025, Japan** — [Poster Presentation](#)
A comparative analysis of the coronal properties of Broad Line and Narrow Line Seyfert 1 galaxies.
- **Astronomical Society of India, Conference 2026, Guwahati** — [Poster Presentation](#)
Nature of Coronae in Bright Active Galactic Nuclei and affecting the surrounding medium.

TECHNICAL SKILLS

High-Energy Astrophysics: Broadband X-ray spectral modelling, timing analysis, X-ray polarization, relativistic reflection, photoionized absorption, multi-epoch AGN spectroscopy, physical model fitting.

X-ray Data Analysis: XSPEC, ISIS, HEASoft, NuSTARDAS, SAS, FTOOLS; calibration, spectral extraction, response generation, background modelling, parameter estimation. **Data Pipelines:** Automated Python pipelines for NuSTAR/XMM-Newton data reduction, spectral and timing products, batch processing, reproducible multi-epoch analysis.

Machine Learning: PyTorch, GPyTorch, Git, Linux, Physics-Informed Neural Networks (PINNs), Deep Gaussian Processes, Siamese, ReFANN, uncertainty-aware deep learning, etc, time-series reconstruction, surrogate modelling for scientific datasets.

Programming: Python (NumPy, SciPy, Astropy, Pandas, Matplotlib), Bash scripting, parallel programming. **Scientific Computing:** Numerical optimisation, Monte Carlo methods, Markov Chain Monte Carlo (MCMC), Bayesian parameter estimation, uncertainty propagation, statistical model comparison.

Astronomical Software: DS9, fv, TOPCAT, Astroquery, HEASARC archive, SDSS database access.

WORKSHOPS

Fast Radio Bursts (FRBs) Workshop

Oct. 6–17, 2025

International Centre for Theoretical Sciences (ICTS), India

Galaxies and Related Topics: Honoring Prof. Biman Nath

Jun. 6–7, 2025

RRI, India

ACHIEVEMENTS AND INTERNSHIPS

- **Internship, NAOJ Tokyo (Summer 2025)** and **collaboration** with Prof. Maria G. Dainotti since April 2024.
- **Full membership, ECAP X-ray Research Group, Germany** since Jan. 2026
- **Research collaboration with the Indian Institute of Astrophysics (IIA), Bengaluru** since Dec. 2024.
- **Management Coordinator, Pravega 2024–25** (India's largest science & cultural festival), IISc